Towards Biophysically-informed Drought Indicators Across US Ecosystems - Figures

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2026-03-29

IMPORTANT: Change "E:/Github/diurnal-divergence-factor/data/" to your own file location in the setup

Then click "Knit" at the tob of the .Rmd file in RStudio

Figure 1. Relative ET VPD plot - Note: this plot was edited in powerpoint after this initial step

Note:

Figure 2 code is in the "Figure_2_REQUIRES_EXTERNAL_FILE.R" script, which requires downloading the aridity index .tif from the original source. -> Information and citation in the script mentioned above.

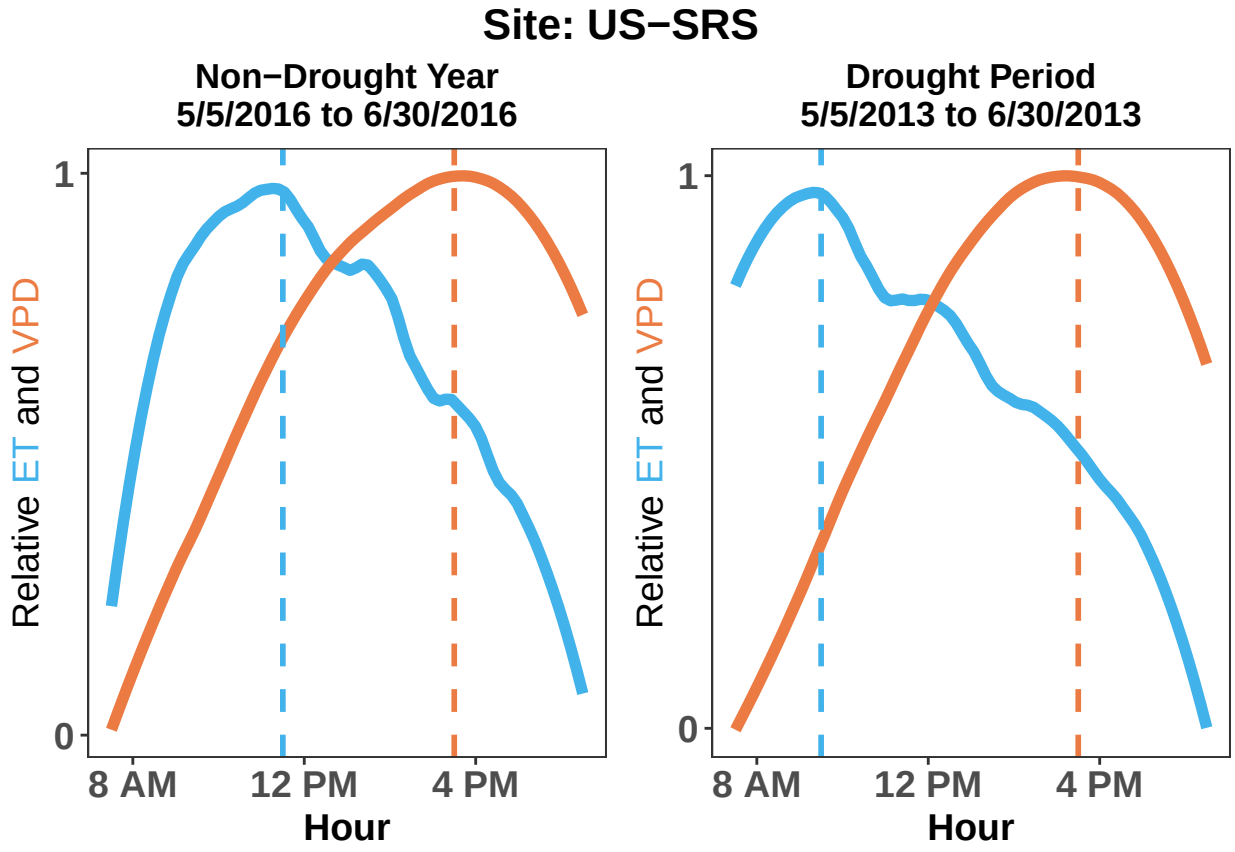


Figure 1: Illustration of the DDF using observed eddy covariance data from a woody savanna site in Arizona (Ameriflux site: US-SRS). Blue depicts ET, with VPD in red, each normalized from 0 to 1 over the day (y-axis) and averaged by hour (x-axis). Panel A shows the typical diurnal sequence during a non-drought period (May-June 2016), with peak ET occurring during midday and peak VPD in the late afternoon. Panel B shows the same time of year during a drought (May-June 2013), where peak ET shifts earlier while VPD timing remains unchanged. This shift illustrates how drought advances peak ET timing relative to peak VPD and forms the conceptual basis for the DDF

Figure 3. Drought versus Non-Drought

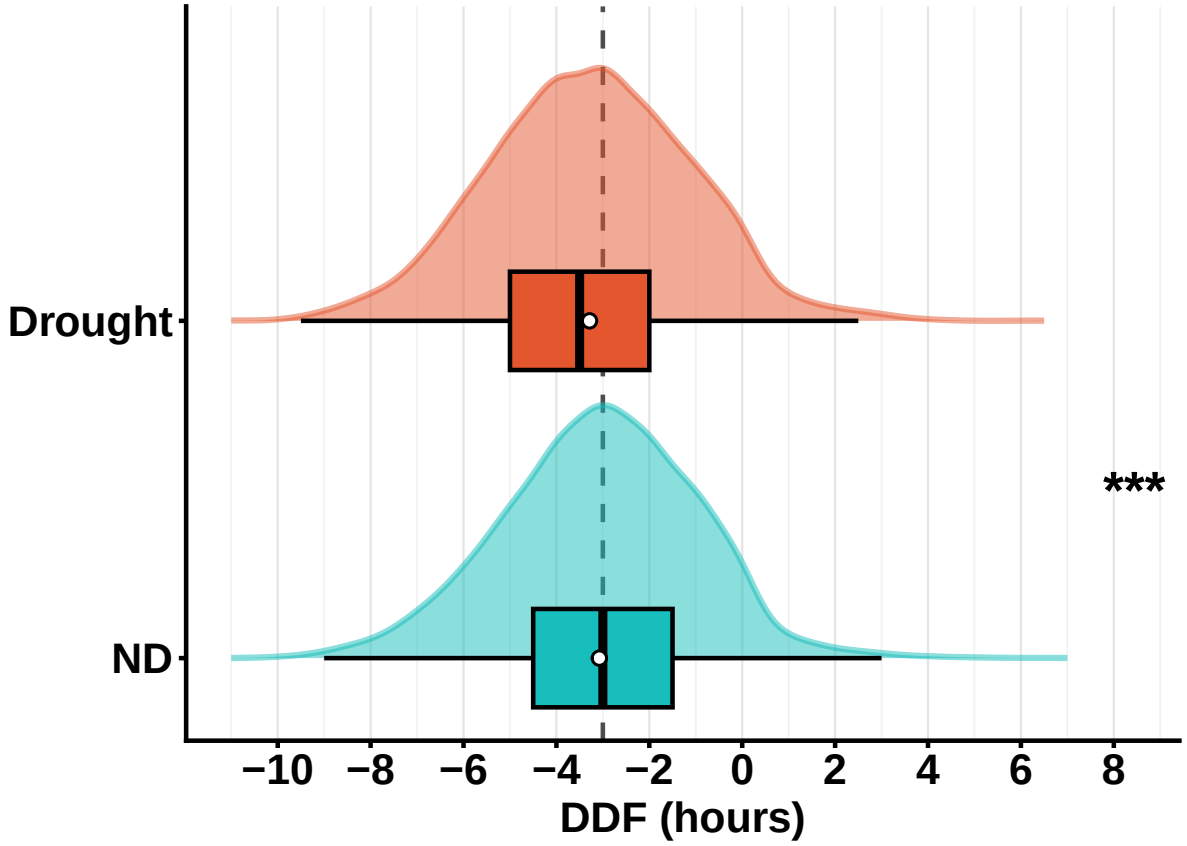


Figure 2: Comparison of DDF distributions across drought vs. non-drought conditions for 40 pooled flux tower sites. Dotted horizontal lines indicate mean ND timing. Boxplots show medians (black horizontal lines) and means (white points) while kernel density overlays (bandwidth = 0.4) further describe the distributions. Statistical significance: A) Welch's t-test (*** $p < 0.001$)

Figure 4. Delta DDF barplots organized by aridity

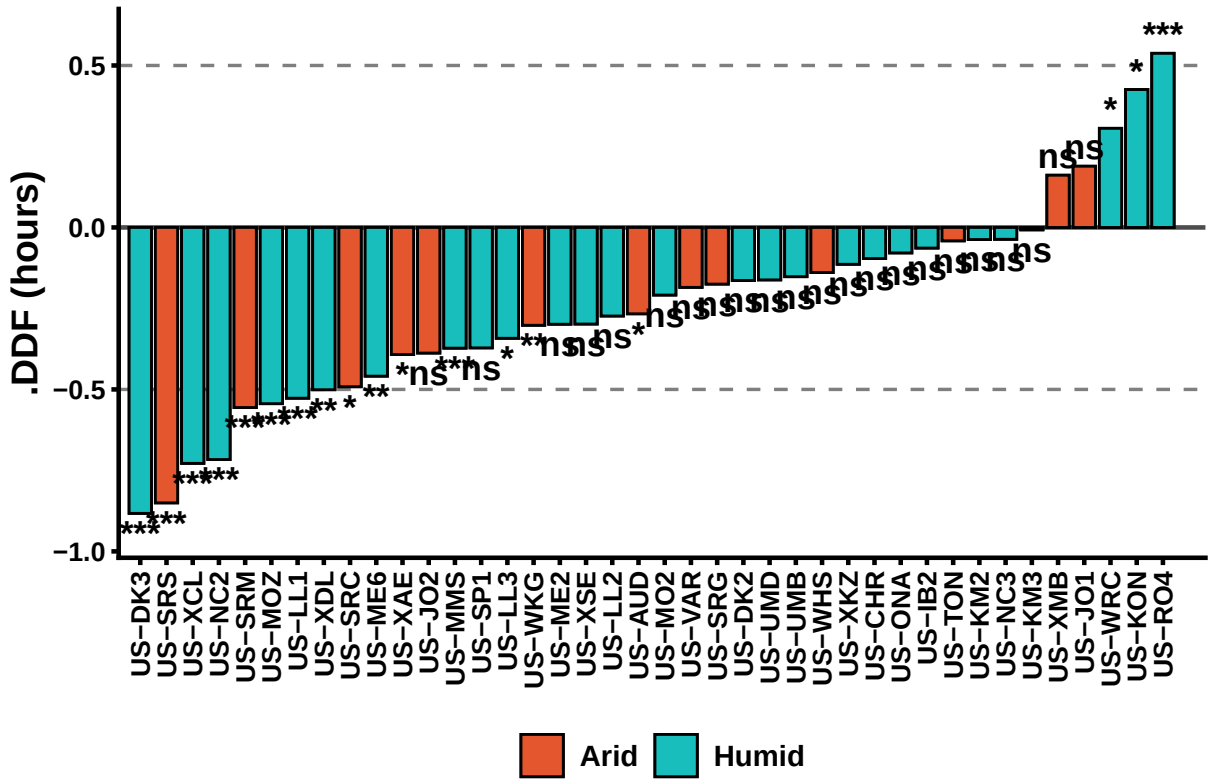


Figure 3: Site-level drought effects on mean DDF (delta DDF = DDFdrought – DDFND), organized by site aridity class then by magnitude of the DDF signal. Negative values indicate earlier ET peak timing during drought

Figure 5. Drought Severity

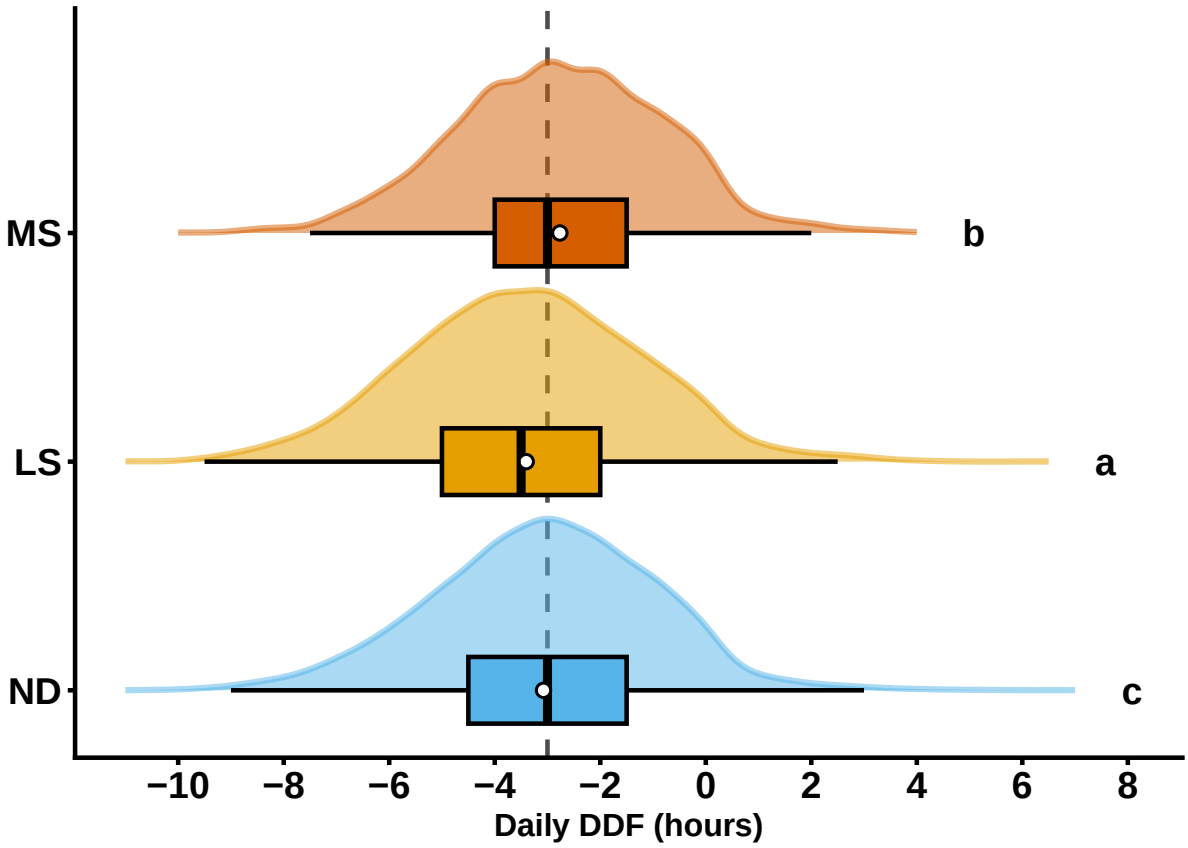


Figure 4: Comparison of DDF distributions across drought severity categories, for 40 pooled flux tower sites. Dotted horizontal lines indicate mean ND timing. Boxplots show medians (black horizontal lines) and means (white points); density overlays Gaussian kernel estimates (bandwidth = 0.4). Statistical significance: Tukey HSD post-hoc comparisons (letters denote statistically significant groups). Outliers were removed for plotting purposes but not analyses

Figure 6. Aridity - Drought vs. non-drought

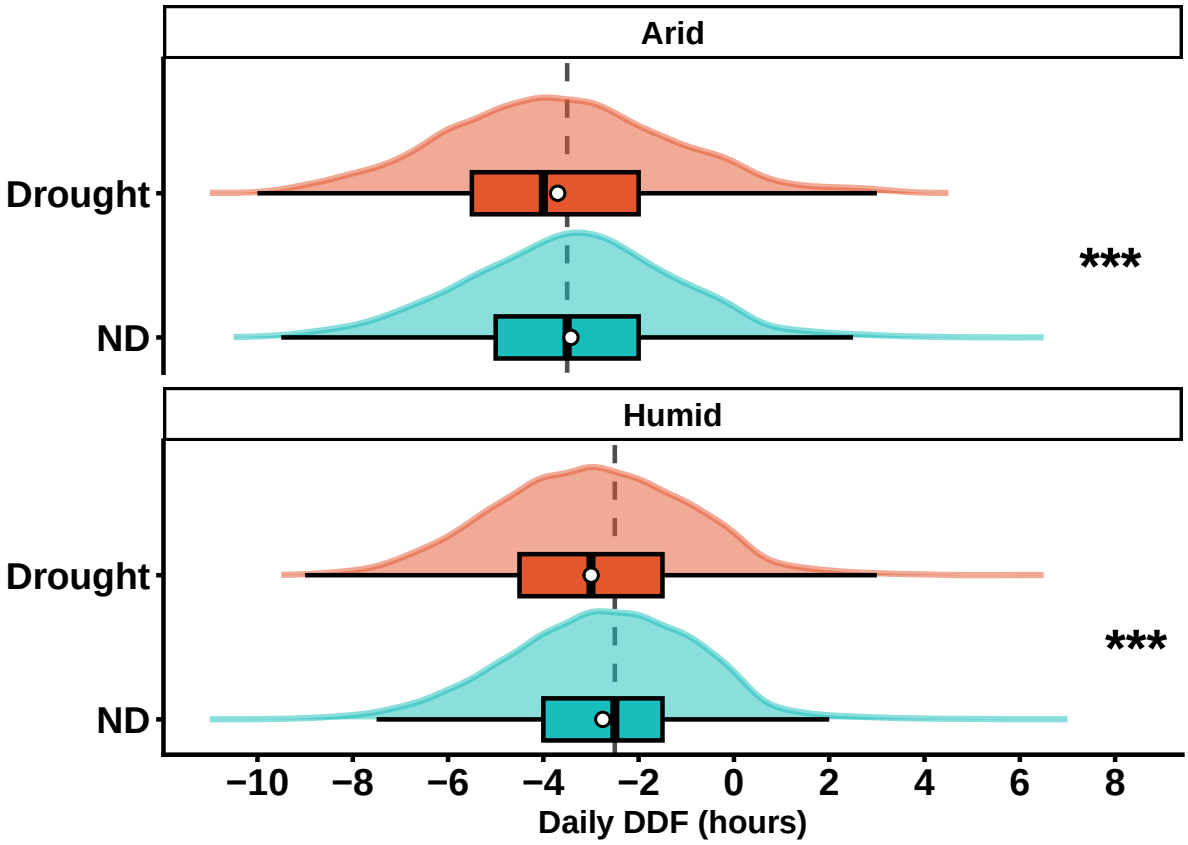


Figure 5: Comparison of DDF distributions across aridity classes during drought and non-drought conditions. Dotted horizontal lines indicate mean ND timing. Boxplots show medians (black horizontal lines) and means (white points); density overlays are Gaussian kernel estimates (bandwidth = 0.4). Asterisks indicate significant differences between drought and ND within each aridity class (Welch's t-test, *** $p < 0.001$). Outliers were removed for plotting purposes but not analyses

Figure 7. Aridity - drought severity

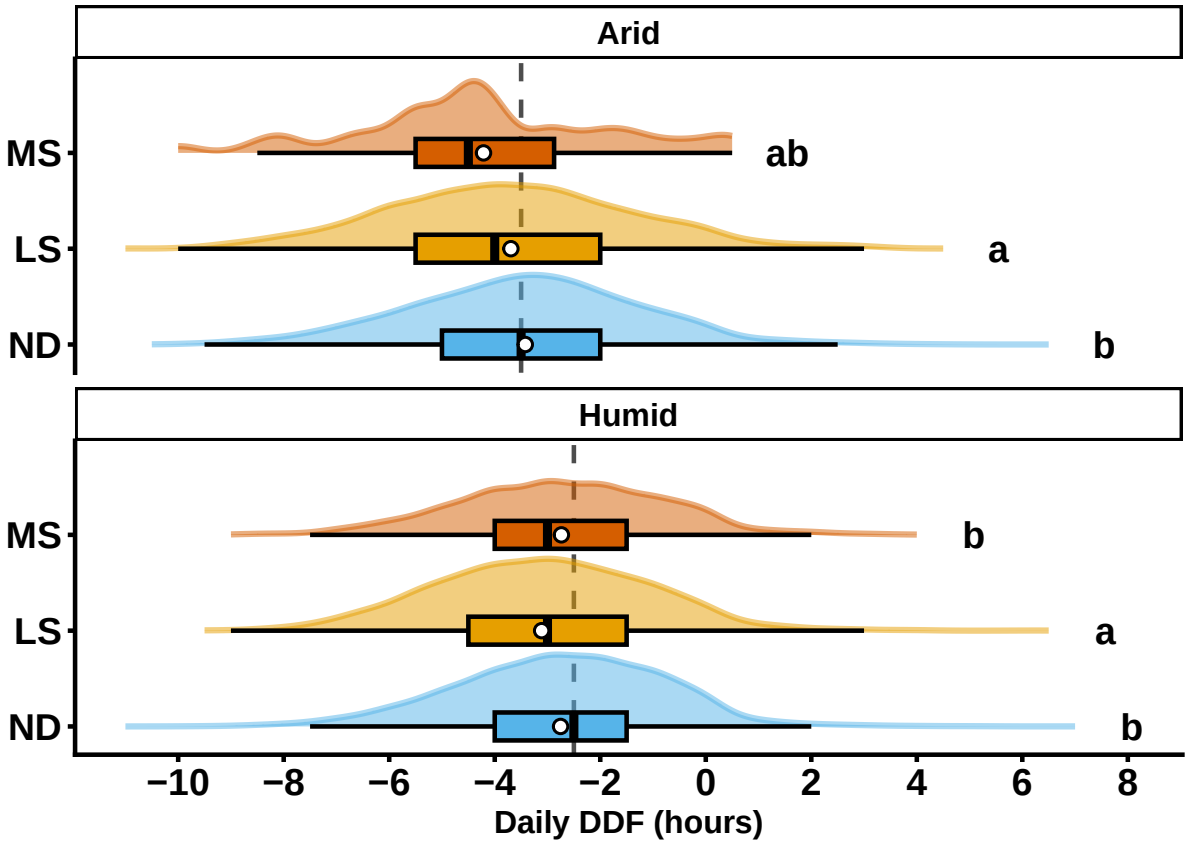


Figure 6: Comparison of DDF distributions across drought severity categories (ND LS, MS) within each aridity class. Dotted horizontal lines indicate mean ND timing for each aridity class. Boxplots show medians (black horizontal lines) and means (white points); density overlays are Gaussian kernel estimates (bandwidth = 0.4). Letters indicate significantly different groups based on Tukey HSD post-hoc tests ($p < 0.05$). Arid sites had no MS droughts detected. Outliers were removed for plotting purposes only

SUPPLEMENTARY PLOTS

Figure S1. Aridity category comparison

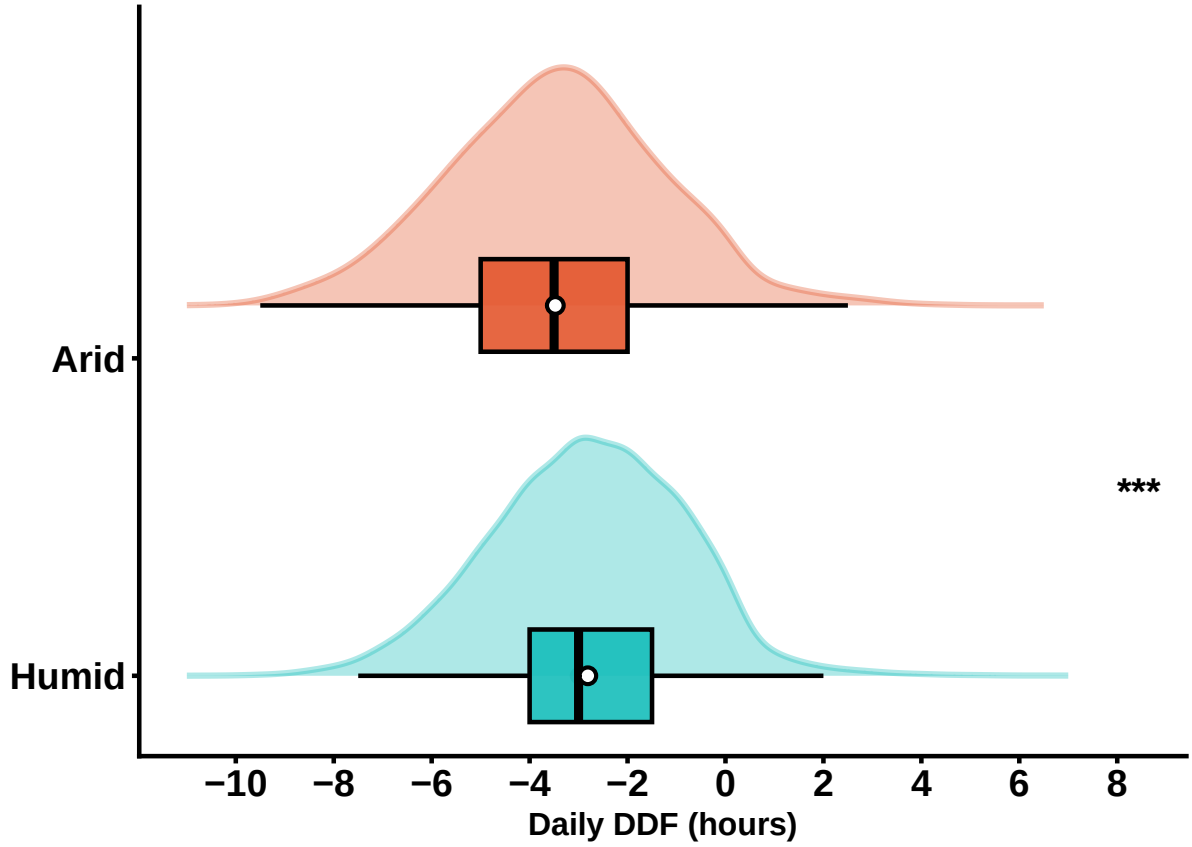


Figure 7: Comparison of DDF distributions across aridity gradients. Boxplots show medians (black horizontal lines) and means (white points); density overlays Gaussian kernel estimates (bandwidth = 0.4). Statistical significance: Welch's t-test (*** p <0.001). Outliers were removed for plotting purposes but not analyses

Figure S2. Vegetation plots

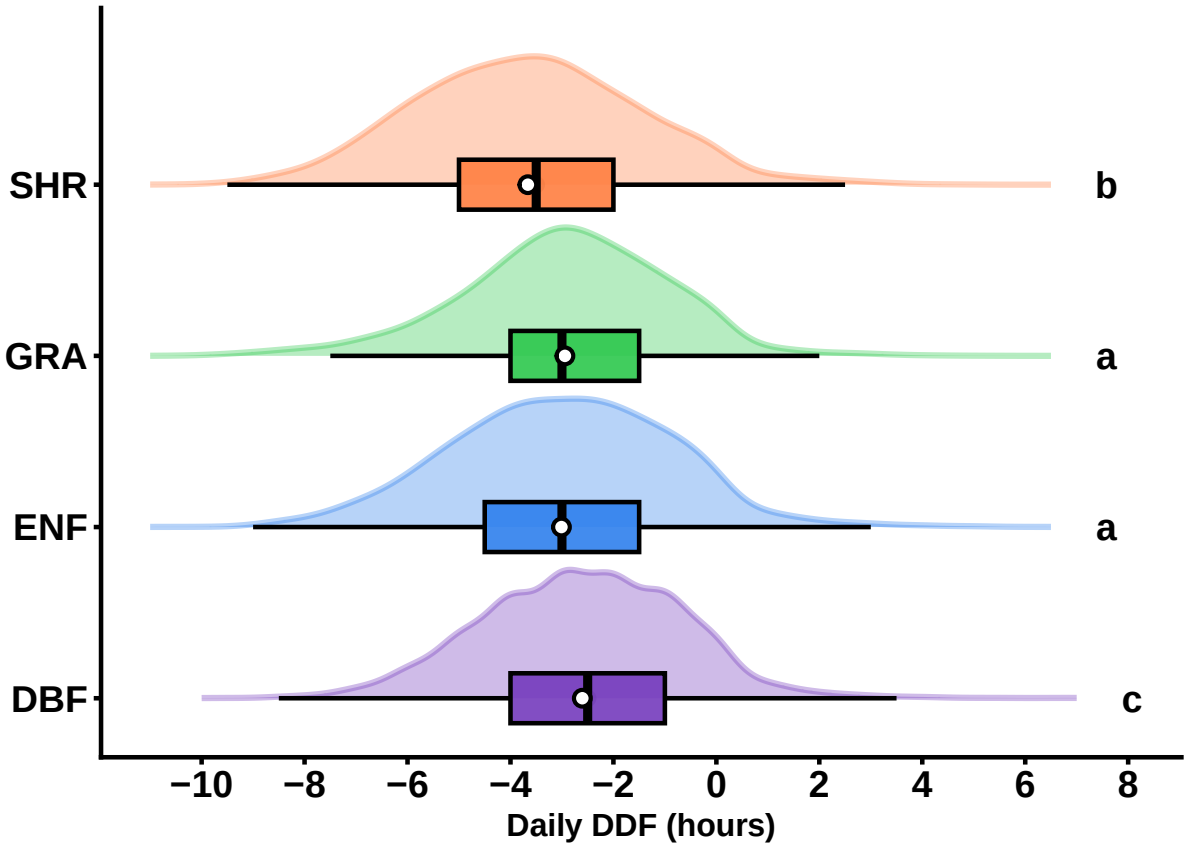


Figure 8: Comparison of DDF distributions across vegetation gradients. Boxplots show medians (black horizontal lines) and means (white points); density overlays Gaussian kernel estimates (bandwidth = 0.4). Statistical significance: Tukey HSD post-hoc comparisons (letters denote statistically significant groups). Outliers were removed for plotting purposes but not analyses

Figure S3. Vegetation drought vs non-drought plots

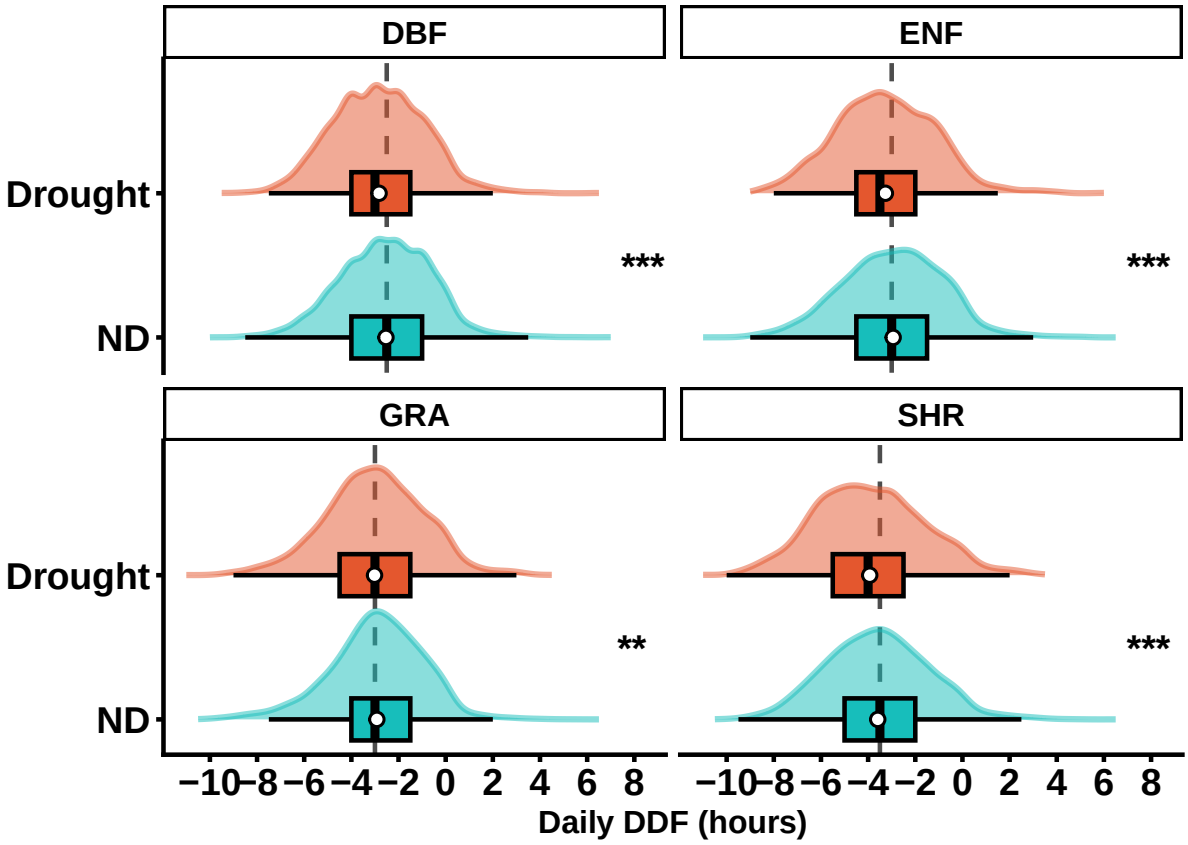


Figure 9: Comparison of DDF distributions and means during drought and non-drought (ND) conditions across Vegetation types. The dotted horizontal line indicates where the timing of peak ET and VPD occur during ND periods for each category on average. Boxplots show medians (black horizontal lines) and means (white points). Half-eye density overlays are unbounded Gaussian kernel density estimates (bandwidth = 0.4) to provide light smoothing for visualization. Significant differences of mean values ($p < 0.05$, *< 0.01*, ***< 0.001***) based on Welch's t-test between groups are indicated by asterisks to the right of the boxplots. Outliers were removed for plotting purposes only

Figure S4. Vegetation drought severity plots

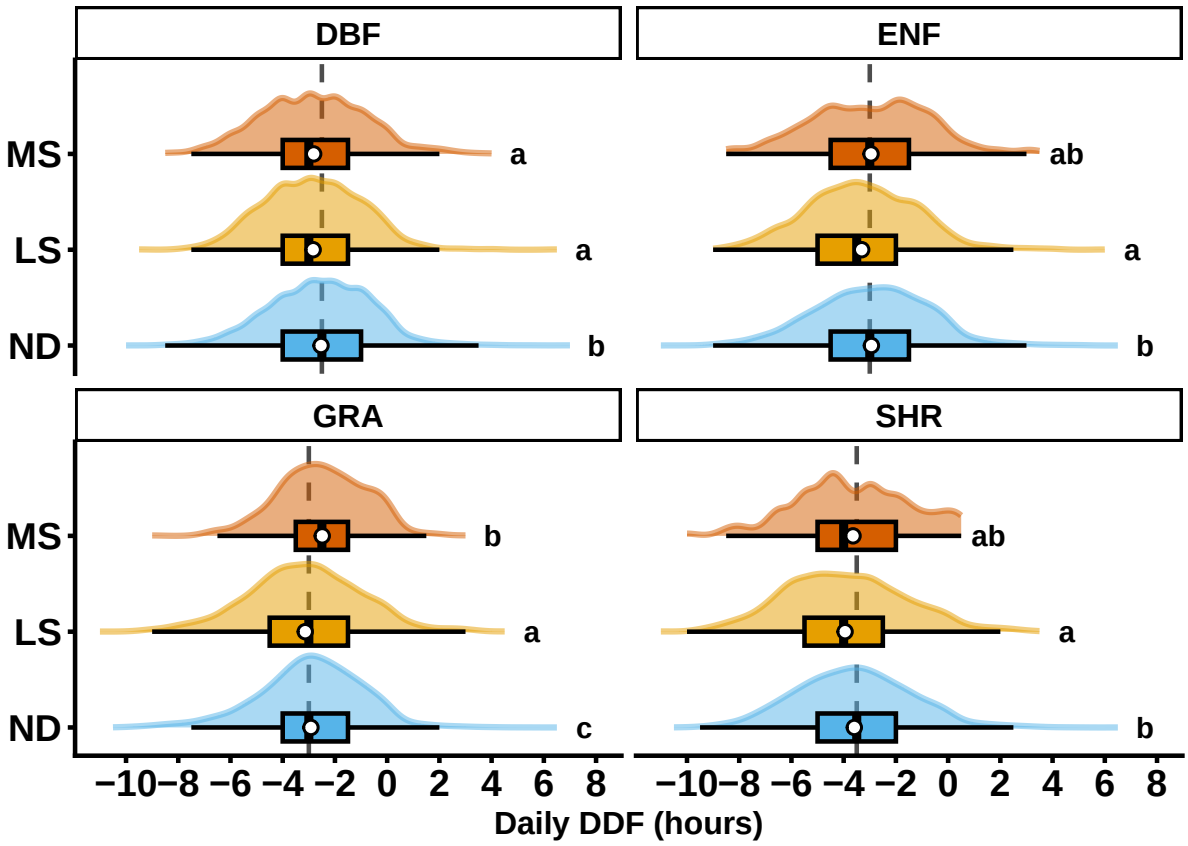


Figure 10: Comparison of DDF distributions and means during each drought severity category across vegetation types. The dotted horizontal line indicates where the timing of peak ET and VPD occur during ND periods for each category on average. Boxplots show medians (black horizontal lines) and means (white points). Half-eye density overlays are unbounded Gaussian kernel density estimates (bandwidth = 0.4) to provide light smoothing for visualization. Letters denote differences in Tukey's post hoc test to the right of the boxplots. Outliers were removed for plotting purposes only. MS sample sizes were small for several vegetation types (ENF: 2 sites; SHR: 3 sites); results should be interpreted with caution